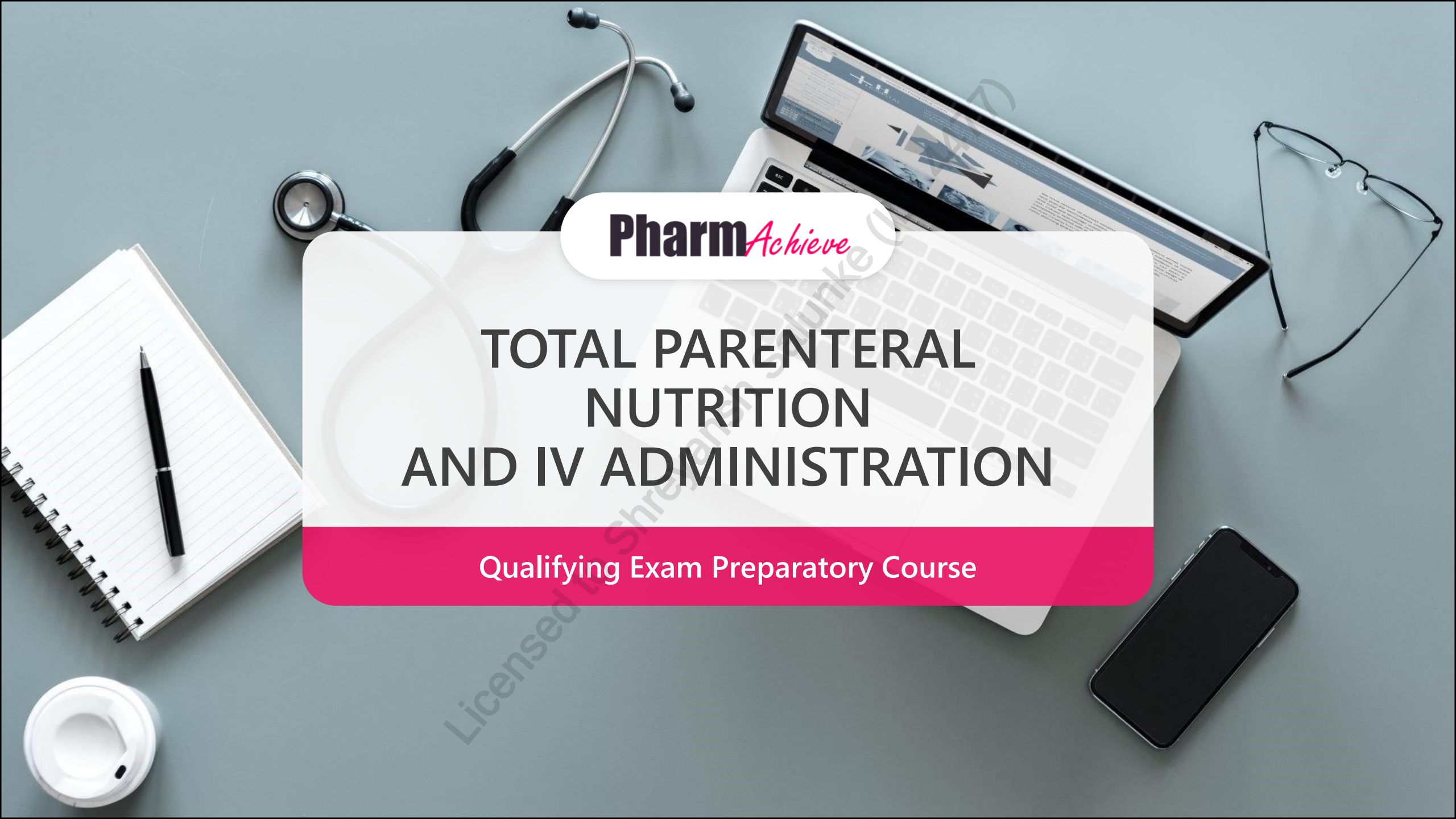


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A central white and pink overlay contains the course title and logo.

Pharm*Achieve*

TOTAL PARENTERAL NUTRITION AND IV ADMINISTRATION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AUD Alcohol Use Disorder

BUN Blood Urea Nitrogen

CBC Complete Blood Count

DOAC Direct Oral Anticoagulant

EN Enteral Nutrition

GIT Gastrointestinal Tract

IV Intravenous

LFT Liver Function Test

NPO Nothing By Mouth

PICC Peripherally Inserted Central Catheter

PO By Mouth

PRN As Needed

PTT Partial Thromboplastin Time

TG Triglycerides

TPN Total Parenteral Nutrition

TABLE OF CONTENTS

- Definitions
- IV Administration
 - Types of IVs
- Total Parenteral Nutrition (TPN)
 - Indications for and contraindications to total parenteral nutrition (TPN)
 - Advantages and disadvantages of TPN
 - Complications associated with TPN
 - Refeeding syndrome
 - Issues with compounding TPN
 - Roles of various healthcare providers

DEFINITIONS

Total Parenteral Nutrition (TPN)

- Method to provide most calories and nutrients that a person needs via **IV route**
- Contains amino acids (proteins), dextrose (carbohydrates), lipids (fats), electrolytes, vitamins and minerals
- Contents of TPN bags (ingredients) are tailored to meet individuals' needs
- Nutrients are delivered intravenously into the bloodstream either centrally (via a central vein) or peripherally (via a smaller vein)
 - **Central TPN**: delivered via subclavian vein through a PICC
For long-term TPN use
 - **Peripheral TPN**: delivered through cephalic or basilic (preferred) vein
For short-term TPN use

Enteral Nutrition (EN)

- Method to deliver nutrition directly into GIT
- Tube is inserted from nose, mouth, stomach or intestine

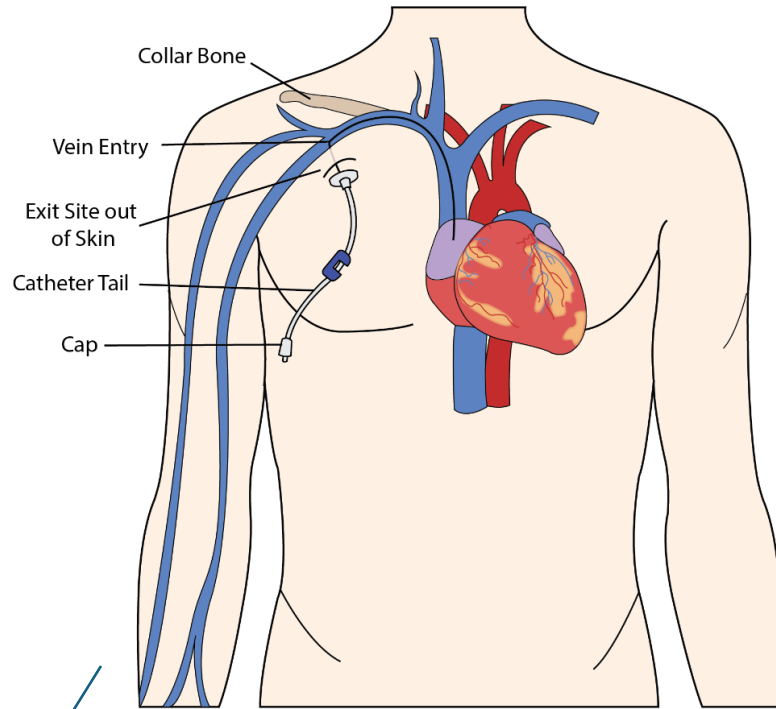
IV ADMINISTRATION

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CENTRAL IV ADMINISTRATION

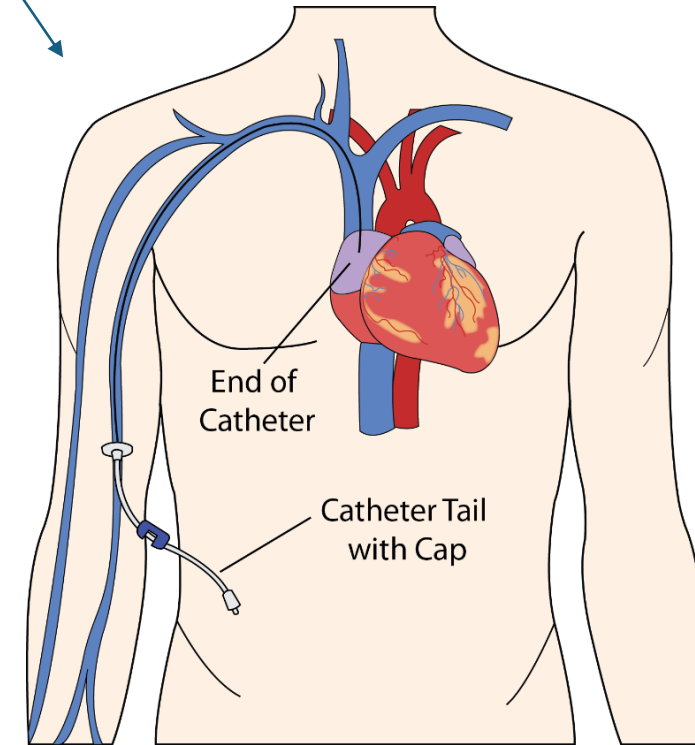


Central Line

A catheter is placed into a large blood vessel (subclavian vein, superior vena cava, etc.) with the tip terminating near the right atrium

Peripherally Inserted Central Catheter (PICC Line)

A type of central catheter where the catheter is placed into the antecubital vein and the tip terminates into or near the right atrium



PERIPHERAL IV ADMINISTRATION

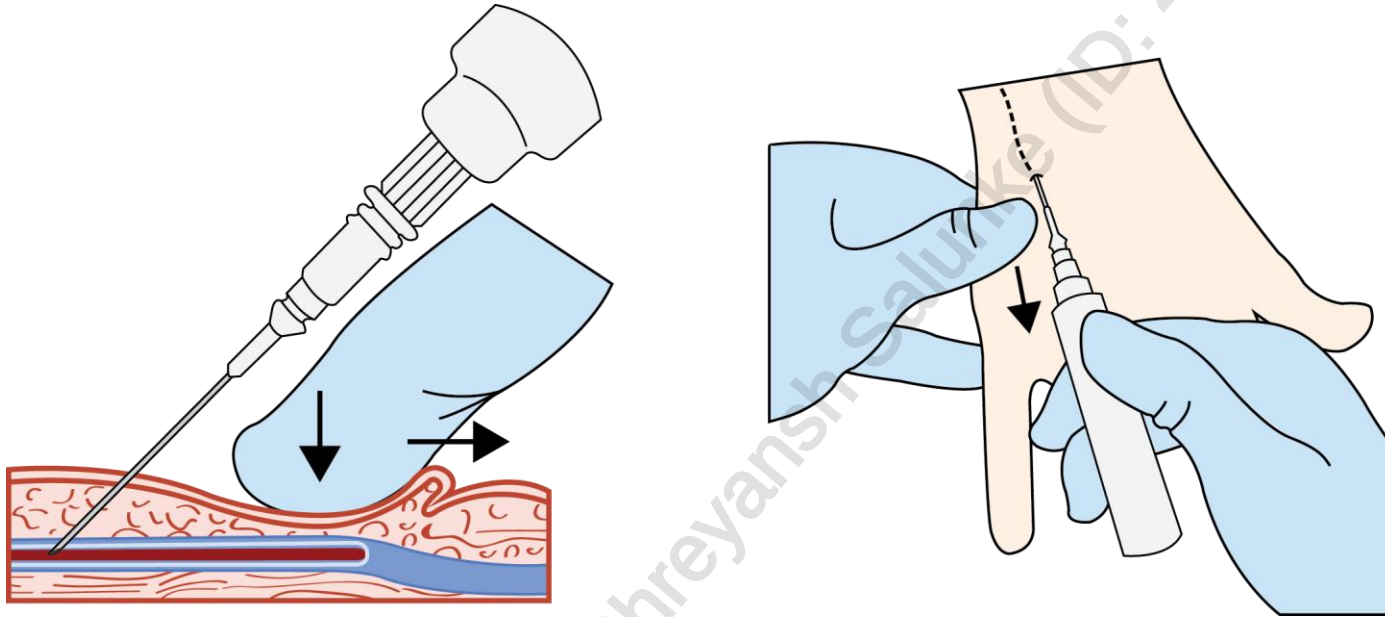


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Peripheral Line

A small catheter (2-15 cm) is placed through the skin into a superficial vein, primarily metacarpal, cephalic, basilic or median vein

TYPES OF IV ADMINISTRATION

Continuous Infusions

- Injected through the primary infusion line
- Examples:
 - Vasopressors
 - Total parenteral nutrition (TPN)
 - Replacement fluids
- Rate: mL/h

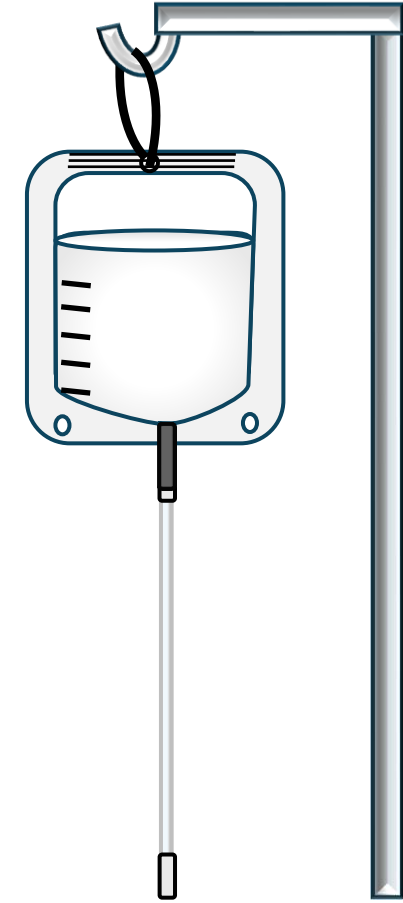


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TYPES OF IV ADMINISTRATION

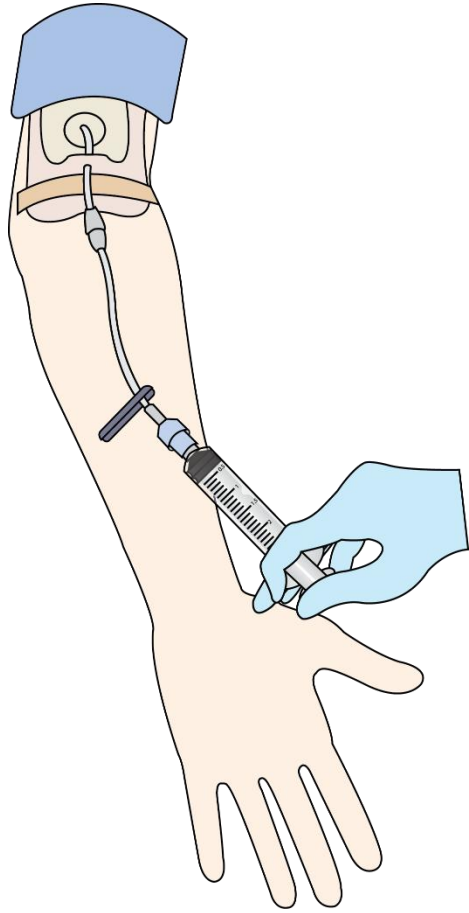


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IV Push

- Intermittent doses (Q12h, Q8h, etc.)
- Bolus that is injected directly into a vein or access port
 - “Immediate peak in concentration”

TYPES OF IV ADMINISTRATION

IV Piggyback

- For medications that need to be diluted or infused slowly
- Intermittent doses (Q12h, Q8h, etc.)
- Secondary infusion
 - Infused through the primary line
- Medications administered include
 - Various antibiotics
 - Concentrated electrolytes (i.e. potassium chloride)

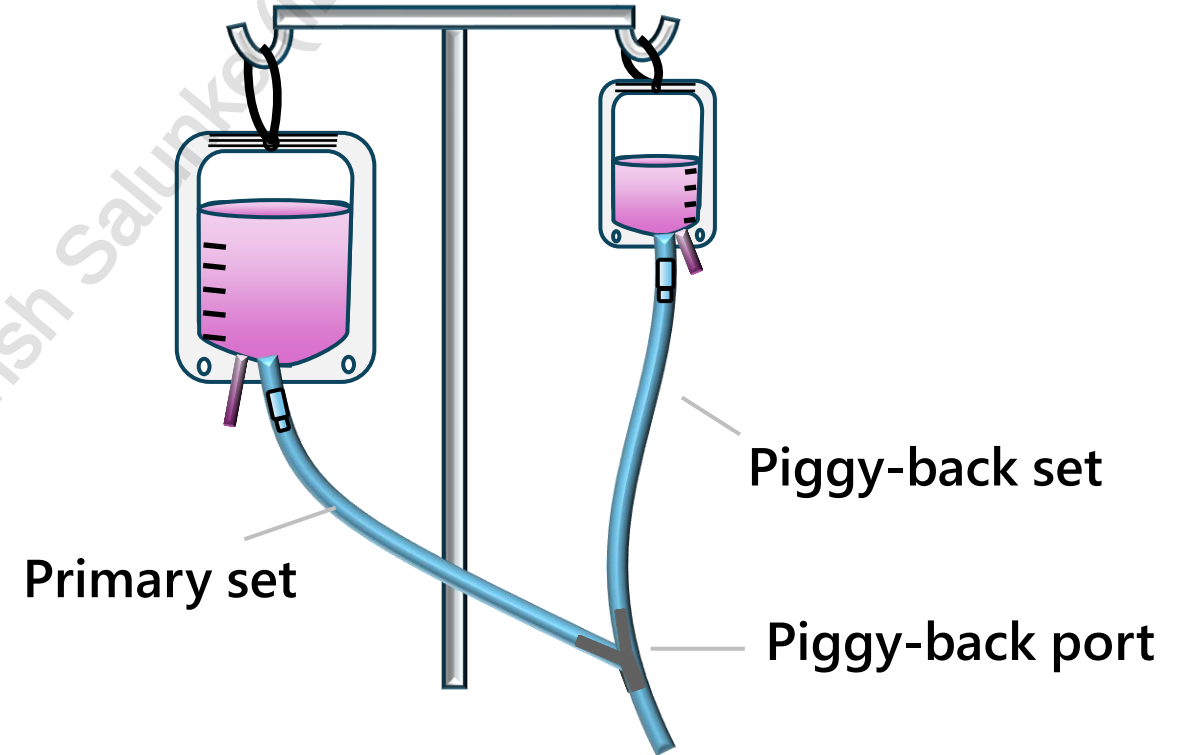


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TOTAL PARENTERAL NUTRITION

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INDICATIONS AND CONTRAINDICATIONS OF TPN

INDICATIONS	CONTRAINDICATIONS
Non-functional/Inaccessible GIT	Functional and accessible GIT
NPO (nothing by mouth) >5-7 days	Feeding in <5 days with no risk of malnutrition
Short bowel syndrome after bowel removal	Prognosis does not warrant aggressive nutritional support (i.e. palliative, terminally-ill)
Mesenteric Ischemia	Inadequate attempts to feed enterally
Severe vomiting/diarrhea >5-7 days	Severe hyperglycemia/electrolyte abnormalities
Small bowel obstruction	Volume overload
Fistula and no distal enteral access	Inadequate IV access
Inadequate/failed EN (major surgery, tube placement)	Risk of TPN outweighs benefit
Severe pancreatitis intolerant of jejunal nutrition	

CENTRAL PARENTERAL NUTRITION

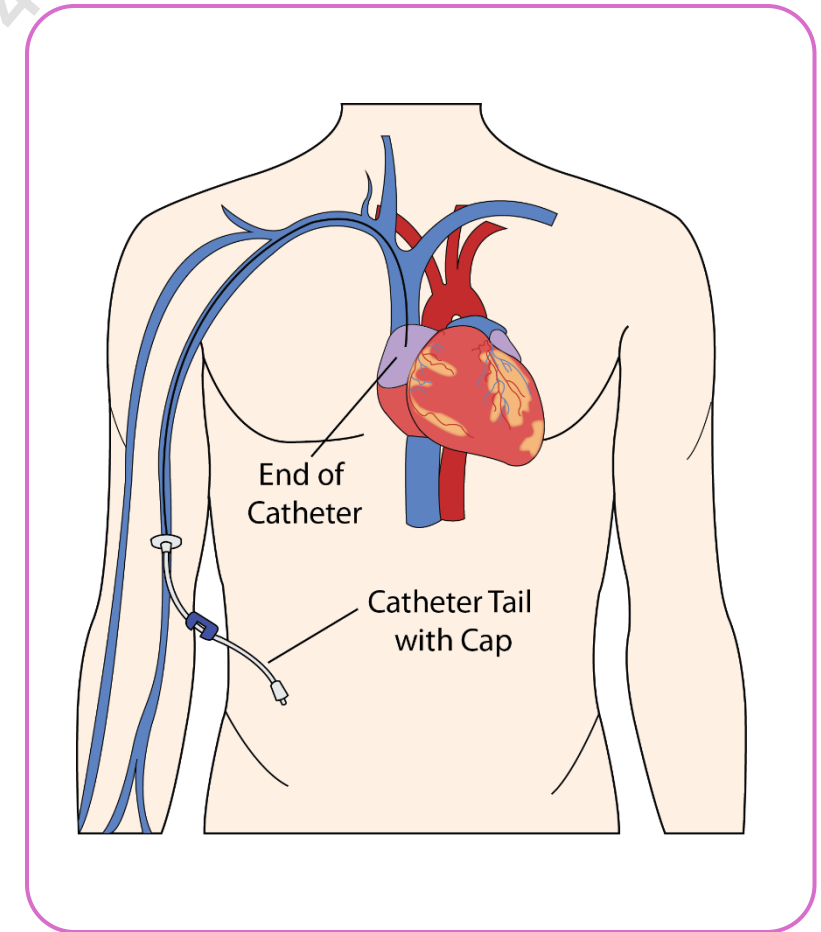
- Delivered via subclavian vein through a PICC
- For long-term TPN use (>14 days)
- Choice of TPN in ambulatory settings

Advantages:

- Infuse larger volumes
- Infuse solutions >900 mOsm/L (high osmolality)
- Give more electrolytes
 - Potassium >80 mmol/L

Disadvantages:

- Requires surgical and radiological procedure to place PICC line
- Risk of infection during access
- Risk of deep vein thrombosis



PERIPHERAL PARENTERAL NUTRITION

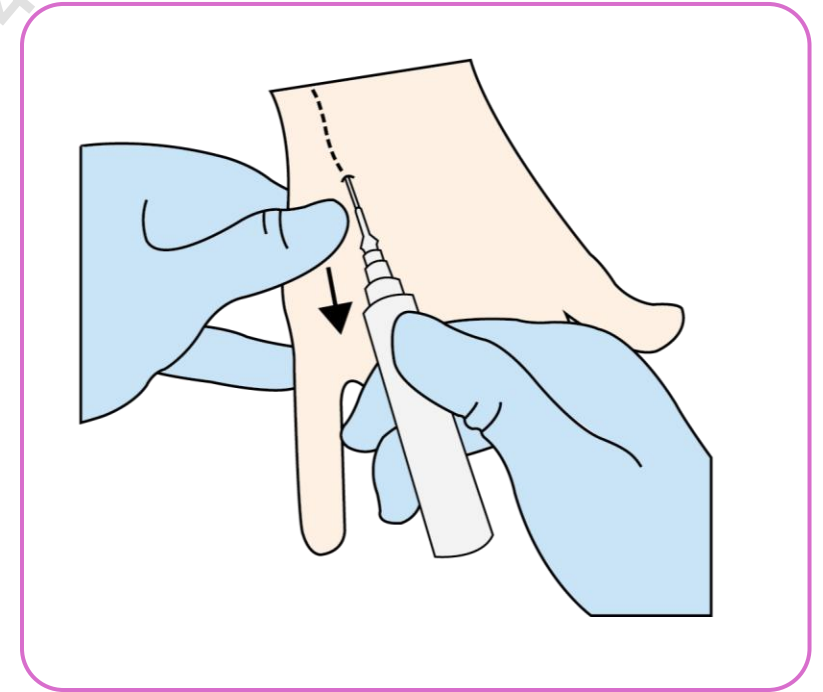
- Delivered through cephalic or basilic (preferred) vein
- For short-term use

Advantages:

- Easy to start due to line being peripheral

Disadvantages:

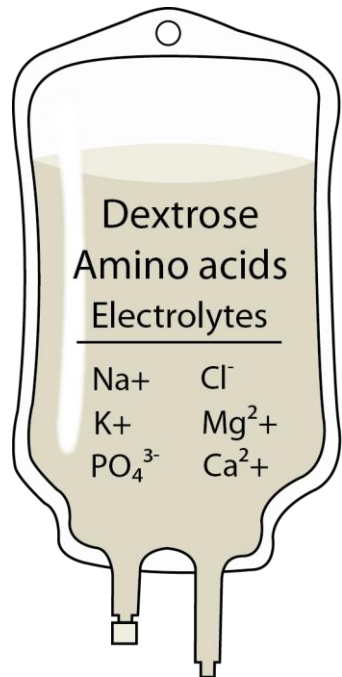
- Need to change line every 48-72 hours
- Smaller volumes only
- Lower osmolality solutions (<900 mOsm/L) limits amounts of nutrition and electrolytes that can be given
- Short-term use <14 days
- Greater risk of thrombophlebitis (blood vessel irritation and clot formation)



TYPES OF TPN

2-In-One TPN

- Dextrose, amino acids and electrolytes are in 1 bag
- Lipids delivered separately



Images: Copyright © PharmAchieve



3-In-One TPN

- Dextrose, amino acids and lipids mixed in 1 bag based on the patient's needs
- Requires a compounding machine (e.g. Micromix®) or use of commercially available multi-chamber bags
- Has a higher osmolality, thus may need to be given centrally

Advantages: ↓ nursing time, easy to administer, ↓ risk of contamination, better fat utilization, continuous infusion of fat

Disadvantages: ↓ compatibility, fat emulsion limits amount of nutrients that can be compounded (or added), limits visual inspection and ↓ ability to detect precipitates

COMPONENTS ADDED TO TPN

Electrolytes and vitamins	Medications	Trace elements
• Potassium • Phosphate • Sodium • Calcium • Magnesium • Multivitamins	• Heparin • Ranitidine • Octreotide • Insulin	• Zinc • Selenium • Copper • Chromium • Manganese

COMPLICATIONS OF TPN

Systemic Infection

- Suspect skin flora (*S. aureus*), *candida*

Mechanical (e.g. catheter obstruction)

- Prevent by flushing carefully
- Treat clots in catheter with alteplase

Venous thrombosis

- DOAC or heparin

Thrombophlebitis

- Due to hyperosmolar solution in peripheral veins

Metabolic

- Renal failure, refeeding syndrome, hyperglycemia, excessive CO₂ production, hypertriglyceridemia, liver dysfunction

REFEEDING SYNDROME

Clinical manifestation

- Refeeding syndrome occurs when nutrition is provided to a patient who is severely malnourished
- Rapid feeding **causes metabolic and hormonal changes**
- Hypophosphatemia (hallmark), hypokalemia, hypomagnesemia, hyperglycemia, respiratory failure

Risk Factors

- If TPN does not contain phosphate
- If a large calorie load given to malnourished patient
- Anorexia or AUD
- Insulin, chemotherapy, antacids, diuretics

Prevention

- Start refeeding at **low rate/amount** in severely malnourished/high risk patients
- **Ensure phosphate is added to TPN** (unless contraindicated in case of hyperphosphatemia in patients with renal dysfunction)

Treatment

- **Replace electrolytes and treat hyperglycemia**
- **Decrease TPN nutrition by up to 50% and increase slowly to nutrition rate goal**

COMMON ISSUES WITH TPN COMPOUNDING

Check for contamination and physical incompatibilities

- Creaming: accumulation of triglycerides (TGs) at the top
- Aggregation: clumping of TGs within emulsion
- Coalescence: fusion of small TGs into larger particles
- Cracking: separation of oil and water components

How to prevent precipitates and incompatibilities:

- Check osmolality of TPN prior to compounding
- Check concentrations of amino acids and electrolytes to be added
- Ensure maximum amounts are not exceeded to avoid precipitation
- Avoid mixing calcium and phosphate in close sequence
- **Add phosphate first** then all the other electrolytes/minerals
- **Add calcium last** to the TPN bag
- Infuse TPN within 24 hours of preparation

MONITORING

	Acute patient	Stable patient
Chemistry (Ca, Mg, PO ₄)	Daily	Weekly
Electrolytes, BUN, Creatinine	Daily	1-3 times weekly
Triglycerides	Weekly or as needed	2-4 times/month or as needed
CBC, PTT	Weekly	Weekly
Glucose	3 times daily if blood sugar increases	3 times daily if blood sugar increases
Weight	Daily	2-3 times weekly
Input and Output	Daily	Daily

Note: It is important to perform baseline measurements for all parameters prior to starting TPN

ROLE OF HEALTHCARE PROVIDERS



Pharmacist

- Distribution: check for compatibility, bag contents, additives
- Clinical: monitor bloodwork, manage complications



Dietitian

- Macronutrient and micronutrient management: calculating amounts of dextrose, amino acids, fats and electrolytes required to meet daily caloric intake



Nurse

- Administer TPN
- Monitor input and output, blood sugar levels, signs of line infection and line occlusion



Physician

- Working with dietitian to assess needs, prescribing TPN, monitoring bloodwork, collaborating with team to help manage complications

TAKEAWAY

- TPN is reserved for critically ill patients with inadequate oral or enteric nutrition therapy
- TPN provides calories and nutrients either centrally for long-term use or peripherally for short-term use
- TPN contains dextrose, amino acids, electrolytes, vitamins, minerals, and trace elements according to patient's nutritional needs
- Complications of TPN include infections, blood clots, and refeeding syndrome
- Routine monitoring includes measurements of input & output, serum electrolytes, glucose, Ca, Mg, PO₄, triglycerides and amino transferases

CASE

RF is a 43-year-old male who was found on the street in a state of severe dehydration and was admitted to the ICU. He presents with signs of muscle reflex loss in the lower extremities and brittle and dry hair. The patient is unable to retain any food or fluids and is now unconscious, with an urgent need for nutritional and fluid replacement. His medical history includes dyslipidemia, AUD (alcohol use disorder), generalized anxiety disorder and depression. He takes rosuvastatin 20mg PO at bedtime, acamprosate 666 mg po TID, lorazepam 1mg PO once daily PRN and sertraline 25mg PO daily. He has no known medication allergies.

1. Given the patient's clinical presentation, medical history, and urgent need for intervention, what is the most appropriate approach to manage his rehydration and nutritional needs in the ICU?
 - a) Oral feeding should always be prioritized over enteral or parenteral support, and Hydralyte® can be used to address the patient's fluid needs
 - b) Oral feeding is not feasible at this time, and enteral support is the next option. Given the signs of malnutrition and the patient's history of AUD, a nasogastric tube may be used to provide 50% of the nutritional goal
 - c) Both oral and enteral feeding are currently not feasible, making total parenteral nutrition (TPN) necessary to meet 100% of the patient's nutritional needs as soon as possible
 - d) Both oral and enteral feeding are limited, and TPN is required. Given the signs of malnutrition and the patient's history of AUD, TPN may provide 50% of the nutritional goal initially



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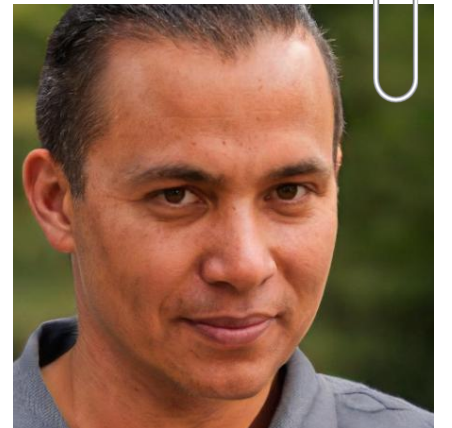
2. What are potential complications of TPN for this patient?
 - a) The patient's medical history of generalized anxiety disorder increases his risk of refeeding syndrome, which may manifest as hypophosphatemia
 - b) The loss of muscle reflexes suggests a state of malnutrition, which could lead to hypokalemia when administering TPN
 - c) Brittle and dry hair are indicative of malnutrition, which may result in hyperkalemia during TPN administration
 - d) The patient's history of alcohol use disorder (AUD) increases his risk of hypoglycemia and potentially respiratory failure



CASE

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3. Which are the correct monitoring parameters for the acutely ill patient like RF?
- a) CBC and triglycerides once daily
 - b) Weight and glucose at least 3 times weekly
 - c) Calcium and BUN once daily
 - d) Input and output twice weekly



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CHANGE LOG

August 2025

- Content reviewed, no edits
- Minor formatting changes, added green text

October 2025

- Updated legend
- Minor formatting changes made throughout

March 2026

- Slide 2: Updated NAPRA competencies
- Slides 14, 15: Minor updates